

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, rates, proportional relationships, and units

04

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

A wind turbine completes 900 revolutions in 50 minutes. At this rate, how many revolutions per minute does this turbine complete?

- A. 18
- B. 850
- C. 950
- D. 1,400

An object's speed is 64 yards per second. What is the object's speed, in feet per second? 1 yard = 3 feet

- A. 61
- B. 67
- C. 94
- D. 192

PROBLEM-SOLVING AND DATA ANALYSIS

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

RATIOS, RATES, PROPORTIONAL RELATIONSHIPS, AND UNITS — SET 4 · PROBLEM-SOLVING AND DATA ANALYSIS 5/31

Q4

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many feet are equivalent to 34 yards? 1 yard = 3 feet

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

How many yards are equivalent to 1,116 inches? 1 yard = 36 inches

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The ratio of the length of line segment XY to the length of line segment ZV is 6 to 1. If the length of line segment XY is 102 inches, what is the length, in inches, of line segment ZV ?

- A. 17
- B. 96
- C. 102
- D. 612

The number of raccoons in a 131-square-mile area is estimated to be 2,358. What is the estimated population density, in raccoons per square mile, of this area?

- A. 18
- B. 131
- C. 149
- D. 2,376

Shaquan has 7 red cards and 28 blue cards. What is the ratio of red cards to blue cards that Shaquan has?

- A. 1 to 4
- B. 4 to 1
- C. 1 to 7
- D. 7 to 1

A cherry pitting machine pits 12 pounds of cherries in 3 minutes. At this rate, how many minutes does it take the machine to pit 96 pounds of cherries?

- A. 8
- B. 15
- C. 24
- D. 36

Q10

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A mechanical device in a workshop produces items at a constant rate of 60 items per hour. At this rate, how many items will the mechanical device produce in 3 hours?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.